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Structurally complex marshes support breeding *Laterallus jamaicensis jamaicensis* (Eastern Black Rail) occupancy

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ABSTRACT

Laterallus jamaicensis jamaicensis (Eastern Black Rail) is a federally threatened, rare, cryptic species whose habitat requirements across the U.S. Gulf Coast remain poorly understood, limiting effective conservation planning and implementation. To enhance our understanding of *L. j. jamaicensis* ecology, we examined the relationship between *L. j. jamaicensis* occupancy and vegetation structure, plant diversity and community composition, and topography. We conducted 2,524 call-broadcast surveys at 281 sites across 25 properties in the U.S. Gulf Coast (coastal Texas, Louisiana, and Florida). Surveys occurred in high marsh and adjacent wetlands (e.g., estuarine marsh) during the *L. j. jamaicensis* breeding season (March–August) in 2022 and 2023 and were paired with annual vegetation surveys. Using stacked single-season occupancy models, we assessed relationships among occupancy, detection covariates, and site-specific characteristics. Detection was highest at warmer temperatures, under low wind conditions, and closer to a new moon. *Laterallus j. jamaicensis* occupied 34% of survey sites, with a predicted detectability of 43%. *Laterallus j. jamaicensis* occupancy was associated with dense canopy cover (>68% mean herbaceous canopy cover), an open, heterogeneous sub-canopy (<48% mean ground stem cover with >15 percentage points of variation), and a heterogeneous surface organic layer (>6 cm of variation in depth). Occupied sites also had consistently short vegetation (16–56 cm visual obstruction height with <23 cm of variation). We observed little to no evidence of consistent differences in vegetation species composition between occupied and unoccupied sites. Together, these results indicate that *L. j. jamaicensis* exhibits consistent structural habitat associations in the study area, despite regional differences in plant communities. These findings underscore the value of structurally complex marshes for *L. j. jamaicensis* populations and provide a foundation for improved monitoring and conservation efforts across the U.S. Gulf Coast.

Keywords: Black Rail, coastal marsh, habitat use, high marsh, *Laterallus jamaicensis*, occupancy, vegetation structure

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LAY SUMMARY

- *Laterallus jamaicensis jamaicensis* (Eastern Black Rail) is a rare, secretive marsh bird that is threatened in the eastern United States, and its conservation is hindered by knowledge gaps about its ecology.
- We surveyed 281 sites in marshes across the U.S. Gulf Coast for *L. j. jamaicensis* and collected data on vegetation structure, plant diversity and species composition, and marsh topography.
- *Laterallus j. jamaicensis* was most likely to occupy sites with dense herbaceous cover, a patchy but open understory, and moderately short vegetation. These conditions are often produced by plants that form tight basal clumps with stems that radiate outward to create a closed canopy over sparse ground cover.
- Habitat structure, including a dense canopy of grasses and rushes covering relatively open ground, increased the likelihood of *L. j. jamaicensis* occupancy, regardless of which plant species were present.
- To support *L. j. jamaicensis*, conservation and restoration efforts should prioritize maintaining or creating structurally complex coastal marsh habitat.

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Las marismas estructuralmente complejas favorecen la ocupación reproductiva de *Laterallus jamaicensis jamaicensis*

RESUMEN

Laterallus jamaicensis jamaicensis es una especie rara, críptica y amenazada a nivel federal, cuyos requerimientos de hábitat a lo largo de la costa del golfo de EUA aún son poco conocidos, lo que limita la planificación y la implementación de medidas efectivas de conservación. Para mejorar nuestra comprensión de la ecología de *L. j. jamaicensis*, examinamos la relación entre su ocupación y la estructura de la vegetación, la diversidad de plantas y la composición de la comunidad, y la topografía. Realizamos 2,524 censos con reproducción de llamadas en 281 sitios distribuidos en 25 propiedades de la costa del golfo de EUA (costa de Texas, Luisiana y Florida). Los censos se realizaron en marismas altas y humedales adyacentes (e.g., marismas de estuarios) durante la temporada reproductiva de *L. j. jamaicensis* (marzo–agosto) de 2022 y 2023, y se complementaron con relevamientos anuales de la vegetación. Mediante modelos de ocupación de una sola temporada apilados, evaluamos las relaciones entre la ocupación, las covariables de detección y las características específicas de cada sitio. La detección fue mayor con temperaturas más cálidas, en condiciones de poco viento y cerca de la luna nueva. *Laterallus j. jamaicensis* ocupó el 34% de los sitios relevados, con una detectabilidad predicha del 43%. La ocupación de *L. j. jamaicensis* estuvo asociada con una cobertura densa del dosel (>68% de cobertura herbácea media del dosel), un estrato subarbóreo abierto y heterogéneo (<48% de cobertura media de tallos a nivel del suelo, con >15 puntos porcentuales de variación) y una capa orgánica superficial heterogénea (>6 cm de variación en profundidad). Los sitios ocupados también presentaron vegetación consistentemente baja (16–56 cm de altura de obstrucción visual, con <23 cm de variación). Encontramos poca o ninguna evidencia de diferencias consistentes en la composición de especies de la vegetación entre los sitios ocupados y no ocupados. En conjunto, estos resultados indican que *L. j. jamaicensis* muestra asociaciones estables con las estructuras de hábitat en el área de estudio, a pesar de las diferencias regionales en las comunidades vegetales. Estos hallazgos destacan el valor de las marismas estructuralmente complejas para las poblaciones de *L. j. jamaicensis* y proporcionan una base para mejorar el monitoreo y los esfuerzos de conservación a lo largo de la costa del golfo de EUA.

Palabras clave: estructura de la vegetación, *Laterallus jamaicensis*, marisma alta, marisma costera, ocupación, uso del hábitat

INTRODUCTION

Coastal wetlands are declining globally due to overexploitation and habitat transformation (Lotze et al. 2006), sea-level rise and increasingly frequent and severe storms (Doney et al. 2012), and altered disturbance regimes (Gedan and Silliman 2009). Salt marshes, a key component of these systems, have experienced particularly severe losses, with an estimated 50%–80% of their global extent lost during the 20th century (Li et al. 2018, Giuliani and Bellucci 2019). In North America, losses have been even greater, with declines of up to 93% compared to pre-settlement conditions (Gedan and Silliman 2009). Salt marshes provide a range of ecosystem services, including coastal protection (King and Lester 1995), erosion control, water purification, carbon sequestration, and fisheries production (Barbier et al. 2011, Li et al. 2018, Barbier 2019). Salt marshes are also among the most productive ecosystems on Earth (Barbier et al. 2011) and provide critical habitat for several breeding bird species, including the U.S. federally threatened *Laterallus jamaicensis jamaicensis* (Eastern Black Rail). Addressing the decline of salt marshes and other coastal wetlands, along with the species that rely on them, is a pressing issue that demands coordinated, multidisciplinary research and deliberate conservation action (Li et al. 2018, Orth et al. 2020).

Laterallus jamaicensis is closely tied to coastal marshes, but substantial knowledge gaps have hindered effective conservation of the species. These gaps are due in part to the species' cryptic behavior and low detectability in dense marsh vegetation, which have limited our ability to accurately monitor distribution and abundance at broad spatial scales. Historically distributed across both coastal and interior wetlands of the eastern United States, the species has experienced widespread declines (Watts 2016). Interior populations have diminished to the point that only a few of unknown size remain (U.S. Fish and Wildlife Service 2019), and roughly 90% of recent breeding season records have occurred in coastal areas, primarily within salt marshes (Watts 2016). Historically, habitat degradation and fragmentation, including the conversion of wetlands to agricultural or developed areas, have driven declines across both coastal and interior populations (U.S. Fish and Wildlife Service 2019, Atlantic Coast Joint Venture 2020). Today, coastal populations continue to be affected by habitat conversion, but additional threats include sea-level rise, flooding, hurricanes, and droughts (U.S. Fish and Wildlife Service 2019, Atlantic Coast Joint Venture 2020), as well as coastal squeeze (where barriers, such as coastal infrastructure, prevent the inland migration of coastal wetlands as sea levels rise; Torio and Chmura 2013) and potential encroachment by black mangrove (*Avicennia germinans*; Armitage et al. 2021).

The scarcity of information on *L. j. jamaicensis* habitat associations and responses to commonly applied wetland management practices has impeded conservation planning, leaving critical knowledge gaps for guiding recovery actions (Atlantic Coast Joint Venture 2020). For example, we know very little about the species' response to prescribed fire, which is commonly applied to coastal marsh systems (Mitchell et al. 2006) and may be a necessary management tool for creating and maintaining *L. j. jamaicensis* habitat (U.S. Fish and Wildlife Service 2019, Atlantic Coast Joint Venture 2020). However, fire may also negatively affect the species by temporarily reducing habitat suitability (Cox et al. 2025), causing nest failure, or resulting in direct mortality (Eddleman et al. 1988, Legare et al. 1998, U.S. Fish and Wildlife Service 2019). Additionally, it is unclear how *L. j. jamaicensis* responds to variation in marsh characteristics across their broad geographic range. Even within the U.S. Gulf Coast, *L. j. jamaicensis* occupies marshes that can vary significantly in structure and plant composition. For example, *L. j. jamaicensis* is associated with *Spartina* spp. (cordgrass) cover in salt marshes (Tolliver et al. 2019, Johnson and Lehman 2021), but they also use *Juncus* spp. (rushes), *Typha* spp. (cattails), and even mixed herbaceous communities in brackish to freshwater marshes (Schwarzer et al. 2024). The use of many different vegetation types across the species' range suggests that management for the species may need to be tailored to local or regional conditions. As such, effective

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conservation depends on understanding how *L. j. jamaicensis* interacts with marsh vegetation within and across their range, where habitat structure, plant species composition, and land management practices vary widely (Atlantic Coast Joint Venture 2020).

Our goal was to evaluate how habitat characteristics influence *L. j. jamaicensis* occupancy in marsh environments across the Gulf Coast of the southern U.S. Specifically, we examined whether occupancy patterns were associated with vegetation structure, plant diversity, or topography in coastal marshes of Texas, Louisiana, and Florida. We also assessed differences in vegetation community composition among states and between occupied and unoccupied survey sites. Together, these analyses addressed a key knowledge gap in *L. j. jamaicensis* conservation by identifying habitat features associated with occupancy across a large portion of its coastal range (Atlantic Coast Joint Venture 2020).

METHODS

Study Area

We conducted this study in marshes along the Gulf Coast of the southern U.S. (hereafter, "U.S. Gulf Coast") in Florida, Alabama, Mississippi, Louisiana, and Texas (Figure 1). The U.S. Gulf Coast is characterized by a humid subtropical climate, with hot, wet summers, mild winters, and frequent tropical storms and hurricanes during summer and autumn. Coastal marshes in this region exhibit varying salinity levels and are characterized by low tidal amplitudes driven by lunar declination (<1 m; Rozas 1995). Hydroperiod, salinity, and nutrient availability are important determinants of vegetation composition across the region (Rozas 1995). Much of our study occurred in or adjacent to high marsh, which occurs at the upper edge of tidal influence and is infrequently flooded by lunar and perigean spring tides, as well as episodic storm events, cold fronts, and wind stress (Rozas 1995). Outside of storms, inundation is shallow (<60 cm), infrequent, and short-lived, typically persisting for less than a day and occurring only a few days per month. Generally, the dominant vegetation species in the habitats we surveyed included *Spartina* and *Juncus* species, including *S. bakeri, S. patens, S. spartinae, J. gerardii*, and *J. roemerianus* (Greenberg et al. 2006). We focused on high marsh and adjacent wetlands (e.g., estuarine marsh) because they are important habitat types for breeding *L. j. jamaicensis* (Tolliver et al. 2019, Johnson and Lehman 2021, Schwarzer et al. 2024) and occur extensively in the region (Greenberg et al. 2006, Enwright et al. 2023).

We conducted this study at 300 sites across 36 properties, including National Wildlife Refuges, state-managed wildlife areas, and private properties. However, our surveys in Alabama and Mississippi yielded too few detections to reliably estimate occupancy (0 detections across 4 sites in Alabama and 3 detections at 1 of 15 sites in Mississippi), and these states were therefore excluded from our analyses to avoid model convergence issues. Our final dataset included surveys from 281 sites across 25 properties in the remaining states. In Florida, we surveyed St. Marks National Wildlife Refuge (30.14°N, –84.24°W), St. Vincent National Wildlife Refuge (29.66°N, –85.13°W), Big Bend Wildlife Management Area (30.05°N, –83.84°W), and adjacent marshes. In Louisiana, we surveyed 13 private properties in Cameron and Calcasieu Parishes, as well as Sabine National Wildlife Refuge (29.90°N, –93.52°W). In Texas, we surveyed McFaddin National Wildlife Refuge (29.67°N, –94.08°W), a private property in Jefferson County, Jocelyn Nungaray National Wildlife Refuge (29.61°N, –94.45°W), Brazoria National Wildlife Refuge (29.14°N, –95.29°W), San Bernard National Wildlife Refuge (28.86°N, –95.54°W), and Big Boggy National Wildlife Refuge (28.76°N, –95.81°W). Each property included 11 ± 13 survey sites (mean ± SD; range: 1–46), spaced at least 500 m apart (Conway 2011). Sites were

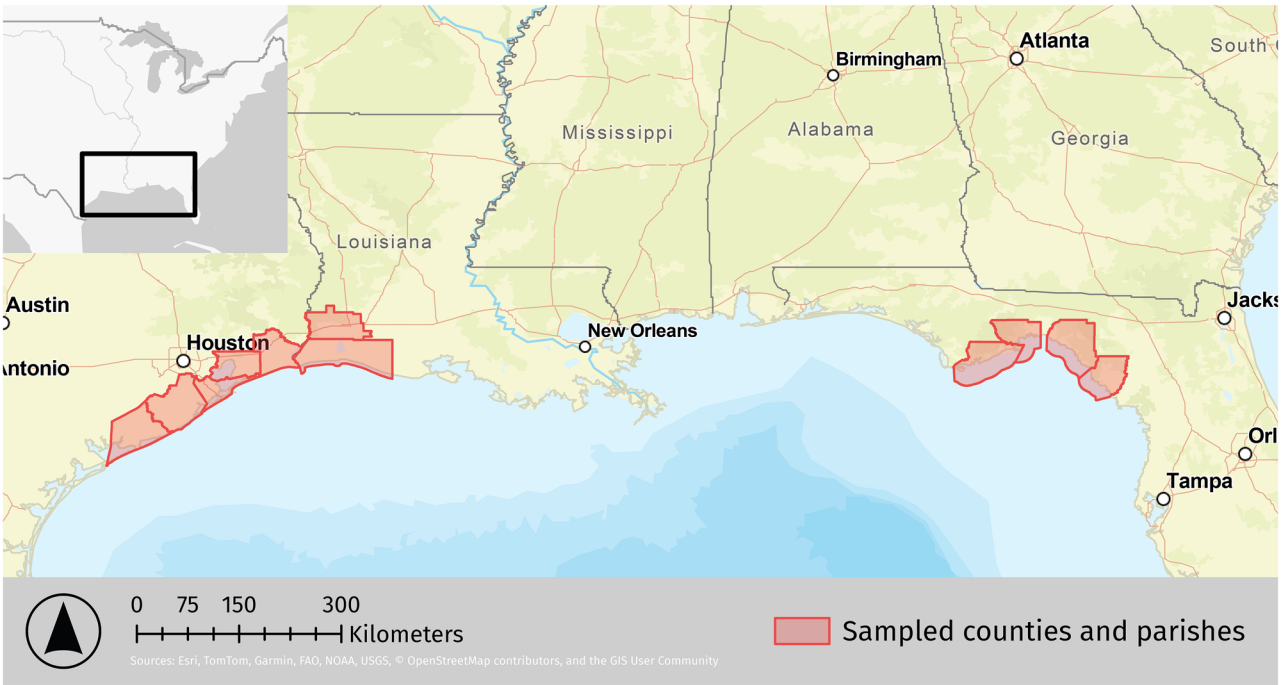


FIGURE 1. Counties and parishes where *Laterallus jamaicensis jamaicensis* (Eastern Black Rail) sampling occurred along the Gulf Coast of the southern United States (inset; upper left) in Texas, Louisiana, and Florida.

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selected based on local knowledge and the high marsh map developed by Enwright et al. (2023). Sites were distributed within wetland boundaries to ensure adequate spacing while maximizing coverage of potential high marsh habitat.

Call-Broadcast Surveys

To detect *L. j. jamaicensis*, we conducted call-broadcast surveys from March to August in 2022 and 2023. This timeframe coincides with the species' breeding season (Hand et al. 2021), when *L. j. jamaicensis* is defending its territories (Legare and Eddleman 2001) and individuals are the most likely to be detected by call-broadcast surveys (Spear et al. 1999, Tolliver et al. 2019, Johnson and Lehman 2021). *Laterallus j. jamaicensis* breeding phenology is poorly documented, but they are thought to start pairing in April and nest from mid-May to mid-August (Eddleman et al. 2020, Watts 2021). We aimed to conduct at least 6 call-broadcast surveys per site per year, distributed across the breeding season and conducted either in the morning (1 hr before to 3 hr after sunrise; range: 0500–1000) or evening (90 min before to 1 hr after sunset; range: 1700–2100). We did not conduct surveys when wind speeds exceeded 25 km hr^{-1} or during sustained precipitation (Conway 2011). Each call-broadcast survey consisted of a structured sequence of alternating listening and broadcasting periods over 10 min: 2 min of passive listening followed by a 30-s broadcast period, repeated 3 times, and concluding with 2.5 min of passive listening. During broadcast periods, we played a series of *L. j. jamaicensis* calls from an NX4 Digital Wildlife Caller (FOXPRO, Lewistown, Pennsylvania, USA). Each broadcast consisted of 10 s of *kickee-doo* vocalizations, 10 s of *kerr-kerr* growl calls, and 10 s of barking *churt* calls (Reynard 1974, Eddleman et al. 2020), all obtained from Xeno-canto (http://www.xeno-canto.org). These are the predominant call types produced by *L. j. jamaicensis* during the breeding season and are widely used in call-broadcast surveys to elicit responses from nearby birds. We do not know the detection distance of human observers listening for *L. j. jamaicensis* calls, but Lamb et al. (2025) demonstrated that the average detection distance of acoustic recording units for 90-dBa simulated *L. j. jamaicensis* calls ranged from 216 to 370 m, depending on call type. We standardized call broadcasts to an amplitude of 80–90 dB at a distance of 1 m in front of the speaker, based on guidelines in the Standardized North American Marsh Bird Monitoring Protocol (Conway, 2011).

During call-broadcast surveys, we recorded whether a *L. j. jamaicensis* was detected (0 or 1). At each survey site, we quantified the survey-specific environmental conditions that can affect the detectability of *L. j. jamaicensis*, including wind speed (km hr^{-1}) and temperature (°C) using a Kestrel 3000 Weather Station, averaged over a 30-s period (Kestrel Instruments, Boothwyn, Pennsylvania, USA). We recorded lunar phase (phases 0–15, ranging from new moon to full moon; https://www.timeanddate.com/moon/phases/) and visually estimated cloud cover to the nearest 10%. We also categorized background noise using a ranking system (0–4) that is commonly applied in marsh bird research (Conway 2011). However, due to limited variability in our surveys, background noise was excluded from subsequent analyses.

Site-Level Characteristics

At each survey site, we quantified several vegetation characteristics, including herbaceous canopy cover, ground-level stem cover, visual obstruction height, surface organic layer depth, species richness, and species evenness (for an overview, see Table 1). We characterized site-level vegetation conditions using 3 nonoverlapping 18-m radius vegetation plots per site, with centers randomly generated within a 250-m buffer around each survey site using a random point generator.

These plots served as the foundation for vegetation sampling, but we added additional plots at sites where *L. j. jamaicensis* were detected. When a *L. j. jamaicensis* was detected during a call-broadcast survey, observers estimated the bird's position and established a vegetation plot at that location. Observers used a laser rangefinder and sighting compass to record the bearing and straight-line distance from their position to the bird's estimated calling location. For each detection, we added 1 plot at the detection location and 3 additional random plots within the site buffer to capture local habitat variability associated with detections.

We conducted vegetation surveys annually from March through August (range: March 8–August 18; mean: June 2; median: June 7). Gulf Coast marsh vegetation typically undergoes rapid spring green-up beginning in February and reaches maturity by May. Surveys occurred when little to no standing water was present, allowing for an accurate measurement of stem cover at ground level. Surveys were adjusted as needed around nesting or brooding *L. j. jamaicensis*, and any plot that risked disturbing active birds was relocated.

Within each 18-m radius plot, we established 6 subsample locations: 4 of them were 10 m from the center in each cardinal direction, one 5 m northeast from the center, and one 15 m southwest from the center. At each of the 6 subsample locations, we measured herbaceous canopy cover, ground-level stem cover, visual obstruction height, and surface organic layer depth. We estimated herbaceous canopy cover within a 50-cm square quadrat by visually estimating the percentage of ground obscured by standing herbaceous vegetation to the nearest 5%. We estimated stem cover separately using a 25-cm quadrat. Observers trimmed vegetation within the quadrat to ~8 cm and then estimated, to the nearest 5%, the percentage of ground obscured by stems. We measured visual obstruction height with a Robel Pole, read from 4 m east of the subsample location at a height of 1 m above the ground. Observers recorded the height of the first visible stripe. Visual obstruction height is highly correlated with vegetation height and aboveground biomass in grasslands and coastal marshes (Robel et al. 1970, Raab et al. 2014). We measured surface organic layer depth with a disc pasture meter (Bransby and Tainton 1977) dropped from a height of 1.5-m above ground. Observers allowed the disc to drop under its own weight, then noted the settling height of the disc. For each variable, we calculated the mean across all subsample locations to represent overall site-level conditions and the standard deviation to capture within-site variability (Table 1).

We assessed species richness and evenness at the 18-m plot level (Table 1). From the plot center, observers visually estimated percent cover (≥5%) for each plant species present. We used these estimates to calculate species richness and Pielou's evenness index (Pielou 1966). We also quantified mean elevation and topographic ruggedness within the 18-m plot using USGS 3DEP Digital Elevation Models (U.S. Geological Survey 2024; Table 1). We calculated mean elevation across the plot directly from these elevation data, and calculated topographic ruggedness by rescaling the Digital Elevation Model to 6-m

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TABLE 1. Summary statistics for site-level covariates included in occupancy models for *Laterallus jamaicensis jamaicensis* (Eastern Black Rail). Table includes a brief description of each covariate, as well as its unscaled mean, standard deviation (SD), minimum, and maximum values across all surveyed sites.

Variable	Description	Mean	SD	Min	Max
Mean canopy cover (%)	Average percentage of ground covered by herbaceous vegetation, based on six 50 × 50 cm quadrat subsamples per plot and ≥3 plots per site. Lower values indicate sparser vegetation cover.	75.9	16.9	12.5	100.0
Mean stem cover (%)	Average percentage of ground covered by ground-level stems, based on six 25 × 25 cm subsamples per plot and ≥3 plots per site. Lower values indicate sparser stem cover.	41.6	17.6	7.5	98.6
Mean visual obstruction (cm)	Average height (cm) at which vegetation obscures visibility of a Robel pole, based on six Robel pole readings per plot and ≥3 plots per site. Lower values indicate less visual obstruction near the ground or shorter vegetation.	46.1	21.7	3.3	183.3
Mean SD canopy cover (%)	Mean standard deviation in percentage of ground covered by herbaceous vegetation, based on six 50 × 50 cm quadrat subsamples per plot and ≥3 plots per site. Reflects within-plot variation in canopy cover; lower values indicate more uniform cover.	16.3	8.1	0.0	40.8
Mean SD stem cover (%)	Mean standard deviation in percentage of ground-level stem cover, based on six 25 × 25 cm quadrat subsamples per plot and ≥3 plots per site. Reflects within-plot variation in stem cover; lower values indicate more uniform stem cover.	19.0	6.5	1.6	43.3
Mean SD visual obstruction height (cm)	Mean standard deviation in the height of visual obstruction, based on six Robel pole readings per plot and ≥3 plots per site. Reflects within-plot variation in vegetation structure; lower values indicate more uniform visual obstruction.	17.2	10.9	1.4	67.3
Mean SD surface organic layer depth (cm)	Mean standard deviation of surface organic layer depth (including litter and herbaceous thatch), based on six pasture disc settling measurements per plot and ≥3 plots per site. Reflects within-plot variation in vegetation and litter thickness; lower values indicate more uniform surface conditions.	9.0	5.8	0.3	51.1
Species richness (count)	Count of unique plant species with >5% cover in the ≥3 18-m radius plots per site.	6.1	2.3	2.0	13.0
Species evenness (0–1)	Pielou's evenness index, calculated from species cover estimates within the ≥3 18-m radius plots per site. Values closer to 1 indicate a more even distribution of cover among species; lower values reflect dominance by one or a few species.	0.8	0.1	0.0	1.0
Elevation (m above sea level)	Mean elevation at site, averaged across ≥3 18-m plots per site. Derived from USGS 3DEP LiDAR data (0.82 meters root mean square error).	1.2	0.6	−0.6	3.5
Topographic Ruggedness Index (m of variation)	Index of elevation variability in the 3 × 3 cell neighborhood surrounding the focal 6-m raster cell at plot center. Averaged across ≥3 18-m plots per site. Higher values indicate more topographic complexity. Method follows Riley et al. (1999).	0.5	0.5	0.0	4.6
State	U.S. state in which the survey site was located: Texas, Louisiana, or Florida.	–	–	–	–

cells and applying the Terrain Ruggedness Index function in Arc Hydro (based on Riley et al. 1999). This index measures variation in elevation among neighboring cells, with higher values indicating more rugged terrain.

We examined all variables for collinearity prior to analysis using Pearson's correlation coefficients. Ultimately, we removed one variable (mean surface organic layer depth) due to high collinearity ($r > 0.60$) with several other variables.

Analyses

We used single-season occupancy models (MacKenzie et al. 2002) implemented with the *unmarked* package (1.5.0; Fiske and Chandler 2011, Kellner et al. 2023) to estimate the probability of *L. j. jamaicensis* occurrence in relation to vegetation characteristics while accounting for imperfect detection. Occupancy models assume that survey sites are closed to demographic processes, such as immigration and emigration, during the sampling period. Because we were explicitly interested in site use by breeding *L. j. jamaicensis*, our sampling protocols focused on regular surveys during a biologically meaningful portion of the year (Valente et al. 2024). We did not sample outside of the breeding season; therefore, our study was not designed to satisfy the closure assumption across survey years. To address this limitation, we used a stacked design that treated each site-year combination as a distinct sampling site and included a site-level random effect to account for the pseudoreplication inherent in fitting two years of data to a single-season occupancy model (e.g., Fuller et al. 2016, Linden et al. 2017, Murray et al. 2021).

To examine how site-level characteristics influenced *L. j. jamaicensis* occupancy, we developed 7 candidate models representing ecologically informed hypotheses (described in Table 2). Models included different combinations of broad categories of site characteristics, including structure (mean herbaceous canopy cover, mean stem density, and mean visual obstruction height), heterogeneity (SD stem density, SD visual obstruction height, and SD surface organic layer depth), diversity (plant species richness and evenness), and topography (elevation and terrain ruggedness) variables. All models that included mean stem density or mean visual obstruction height also included a quadratic term to allow for potential non-linearity in responses. All models included wind speed, temperature, lunar phase, hour of day, and cloud cover as detection covariates, along with state and year as occupancy covariates and a site-level random effect (intercept only). We compared candidate models using Akaike's Information Criterion corrected for small sample size (AIC_c) within an information-theoretic framework (Burnham and Anderson 2002). Within the best-supported models ($\Delta AIC_c \leq 2$), we examined the relative importance of variables by comparing parameter estimates, which were scaled to facilitate comparisons. We evaluated importance using 85% confidence intervals (CIs) (Arnold 2010, Sutherland et al. 2023); covariates were considered influential when their CIs did not overlap zero.

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TABLE 2. Model selection results for single-season occupancy models evaluating relationships between vegetation structure and *Laterallus jamaicensis jamaicensis* (Eastern Black Rail) presence. Models are ranked by Akaike's Information Criterion corrected for small sample size (AIC_c). For each model, we report the number of estimated parameters (k), negative log-likelihood (neglogLik), ΔAIC_c (difference from the best model), and AIC_c weights (w_i). All models included wind speed, temperature, lunar phase, hour of day, and cloud cover as detection covariates, along with state and year as occupancy covariates and a site-level random effect (intercept only). All models that include mean stem density or mean visual obstruction height also include a quadratic term to allow for potential non-linearity in responses.

Model*	Model description	k	neglogLik	ΔAIC_c	w_i
P, ψ (mean canopy cover + mean stem density + mean stem density^2 + mean visual obstruction + mean visual obstruction^2 + SD stem density + SD visual obstruction + SD surface organic layer depth)	Vegetation structure and heterogeneity model: Black rail occupancy is affected by vegetation structure and structural heterogeneity	18	632.82	0.00	0.75
P, ψ (mean canopy cover + mean stem density + mean stem density^2 + mean visual obstruction + mean visual obstruction^2 + SD stem density + SD visual obstruction + SD surface organic layer depth + richness + evenness)	Full vegetation model: Black rail occupancy is affected by vegetation structure, structural heterogeneity, and diversity	20	632.09	2.53	0.21
P, ψ (mean canopy cover + mean stem density + mean stem density^2 + mean visual obstruction + mean visual obstruction^2 + SD stem density + SD visual obstruction + SD surface organic layer depth + richness + evenness + elevation + ruggedness)	Global model: Black rail occupancy is affected by vegetation structure, structural heterogeneity, diversity, and topography.	22	631.65	5.65	0.04
P, ψ (mean canopy cover + mean stem density + mean stem density^2 + mean visual obstruction + mean visual obstruction^2)	Vegetation structure model: Black rail occupancy is affected by vegetation structure	15	648.80	25.94	0.00
P, ψ (richness + evenness)	Vegetation diversity model: Black rail occupancy is affected by vegetation diversity	12	673.19	68.74	0.00
P, ψ (null)	Null model: Black rail occupancy is not affected by measured characteristics.	10	675.98	70.32	0.00
P, ψ (elevation + ruggedness)	Topography model: Black rail occupancy is affected by site topography.	12	675.22	72.80	0.00

*P denotes the detection model and ψ the occupancy model. SD = standard deviation.

We used non-metric multidimensional scaling (NMDS; Kruskal 1964) to explore patterns in vegetation species composition across sites using a species abundance matrix and a Bray–Curtis distance index (Bray and Curtis 1957), considering ordinations unreliable when stress exceeded 0.20 (Clarke 1993). We then conducted permutational multivariate analysis of variance (PERMANOVA) with 999 permutations using the *vegan* package (2.6.10; Oksanen et al. 2025) to test whether vegetation composition differed among states (Florida, Texas, and Louisiana) and, separately within each state, between sites with and without *L. j. jamaicensis* detections. To ensure that observed differences were not driven by unequal within-group variation, we tested for homogeneity of multivariate dispersion via permutation tests on the distances to group centroids (Anderson 2006). To identify which plant species contributed most to differences in community composition among states, we used the *envfit* function in the *vegan* package 2.6.10 (Oksanen et al. 2025). Species scores were fitted onto the NMDS ordination, and vectors with significant correlations (permutation test, 999 permutations) were interpreted as contributing to variation in ordination space among Texas, Louisiana, and Florida. To focus on the most influential species, we applied a conservative significance threshold of $P \leq 0.001$, thereby limiting interpretation to plant species that made highly significant contributions to variation in ordination space.

All analyses were performed in R 4.5.0 (R Core Team 2025) and RStudio 2025.05.0 (RStudio Team 2025). We processed spatial data in ArcGIS Pro 3.4.3 (Esri 2024).

RESULTS

We surveyed a total of 475 site-years (hereafter, "sites"), visiting each site 5.3 ± 2.4 times (mean ± SD; range: 1–11), for a total of 2,524 surveys. Accounting for imperfect detection, *L. j. jamaicensis* occupied 34% of sites (85% CI: 30.9–37.6%), and the estimated probability of detecting a *L. j. jamaicensis* during a single survey at an occupied site was 43% (85% CI: 39.9–45.8%).

L. j. jamaicensis was more likely to be detected during warmer temperatures (β = 0.6 ± 0.1, 85% CI: 0.4–0.7), calmer wind conditions (β = –0.5 ± 0.1, 85% CI: –0.6 to –0.3), and closer to a new moon (β = –0.2 ± 0.1, 85% CI: –0.3 to –0.1; Table 3, Figure 2). At average survey conditions, detection probability increased by 3.1 percentage points for every 1°C increase in temperature (Figure 2A). In contrast, detection probability decreased by 2.2 percentage points for every 1 km hr^{-1} increase in wind speed (Figure 2B), and 1.2 percentage points for each unit increase in lunar phase (Figure 2C). We found little evidence that time of day or cloud cover affected the detectability of *L. j. jamaicensis* across our surveys.

Our best-supported occupancy model included all vegetation structure and heterogeneity covariates but excluded diversity and topography variables, accounting for 75% of the total model weight (w_i; Table 2). The second-ranked model added diversity variables and produced a slight improvement in model fit (negative log-likelihood = 632.1 vs. 632.8), but the increase in complexity resulted in a higher ΔAIC_c (2.5, $w_i = 0.2$). The top model was ~3.6 times more supported by the data, indicating that adding vegetation diversity provided little additional explanatory power. All other models had $\Delta AIC_c > 5.6$ and negligible support. Collectively, these results suggest that *L. j. jamaicensis* occupancy at our study sites was primarily influenced by vegetation structure and heterogeneity rather than vegetation diversity or site topography.

Within the best model, the strongest continuous predictors of *L. j. jamaicensis* occupancy, in order of decreasing effect

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size, were: U.S. state, mean ground-level stem cover, heterogeneity in surface organic layer depth, mean herbaceous canopy cover, heterogeneity in visual obstruction height, heterogeneity in ground-level stem cover, and mean visual obstruction height (Table 3; Figure 3). We found no clear relationship between *L. j. jamaicensis* occupancy and year (Table 3).

L. j. jamaicensis occupancy varied among states and was significantly higher in Florida (β = 1.7 ± 0.8, 85% CI: 0.6–2.8) and Texas (β = 3.0 ± 0.6, 85% CI: 2.1–3.8), relative to Louisiana (Table 3). Model-predicted occupancy under average conditions was 60.7% in Texas (85% CI: 45.4–71.8), 30.3% in Florida (85% CI: 13.4–54.9), and 7.5% in Louisiana (85% CI: 3.7–14.4). *Laterallus j. jamaicensis* occupancy probability was generally highest at sites with lower mean ground-level stem cover (β = −1.2 ± 0.3, 85% CI: −1.7 to −0.7), greater heterogeneity in surface organic layer depth (β = 1.0 ± 0.3, 85% CI: 0.5–1.5), higher mean herbaceous canopy cover (β = 1.0 ± 0.3, 85% CI: 0.6–1.3), lower heterogeneity in visual obstruction height (β = −0.8 ± 0.3, 85% CI: −1.3 to −0.4), greater heterogeneity in ground-level stem cover (β = 0.7 ± 0.2, 85% CI: 0.4–1.1), and intermediate visual obstruction heights, with occupancy declining at both lower and higher values (β = −0.3 ± 0.3, 85% CI: −0.7–0.2; β^2 = −0.4 ± 0.2, 85% CI: −0.6 to −0.2; Table 3, Figure 3). *Laterallus j. jamaicensis* occupancy exceeded 50% when sites had <48% mean ground-level stem cover, >6.4 cm of variation in surface organic layer depth, >68% mean herbaceous canopy cover, <23 cm of variation in visual obstruction height, >15 percentage points of variation in ground-level stem cover, and 16–56 cm of mean visual obstruction height (Figure 3).

Vegetation community composition varied across the U.S. Gulf Coast, with significant differences in vegetation communities among Texas, Louisiana, and Florida ($F_{2,\ 458}$ = 16.90, R^2 = 0.07, P < 0.001). Louisiana sites clustered in ordination space along vectors associated with *Distichlis spicata* (seashore saltgrass) and *Juncus roemerianus* (black needlerush), whereas Florida sites clustered along vectors associated with *Spartina spartinae* (Gulf cordgrass), *Schoenoplectus pungens* (common threesquare), *Stenotaphrum secundatum* (St. Augustine grass), and *Ambrosia psilostachya* (Western ragweed). The vector for *Spartina patens* (salt-meadow cordgrass) separated sites in Florida and Louisiana from those in Texas. Texas sites occupied a broader region of ordination space and aligned with several additional species, including native salt-marsh grasses, such as *Spartina pectinata* (prairie cordgrass), *Sporobolus virginicus* (coastal dropseed), *Juncus gerardii* (salt-meadow rush), and *Cyperus echinatus* (globe flatsedge); legumes including *Desmanthus illinoensis* (prairie mimosa) and *Mimosa nuttallii* (Nuttall's sensitive-briar); and non-native or invasive species, such as *Alternanthera philoxeroides* (alligator weed), *Oxalis dillenii* (slender yellow woodsorrell), and *Euphorbia cyparissias* (cypress spurge). These species exhibited the strongest correlations with the NMDS configuration ($P \leq 0.001$), indicating they were strongly aligned with the dominant compositional gradients represented in the ordination.

TABLE 3. Parameter estimates, standard errors, and 85% confidence intervals (model-generated) for parameters in the top-ranked single-season occupancy model for *Laterallus jamaicensis jamaicensis* (Eastern Black Rail) in salt marsh habitat across the Gulf Coast of the southern U.S. Parameters for which 85% confidence intervals did not overlap zero are shown in bold. Parameters are ordered by linear effect size. All continuous covariates were scaled prior to analysis to facilitate comparison among parameters.

			85% confidence level	
Parameter	**Estimate**	**SE**	**Lower**	**Upper**
Fixed effects (ψ)				
Intercept	**−2.30**	**0.50**	**−3.01**	**−1.57**
State (Texas vs. Louisiana)	**2.95**	**0.59**	**2.10**	**3.80**
State (Florida vs. Louisiana)	**1.68**	**0.75**	**0.60**	**2.76**
Mean stem cover	**−1.20**	**0.32**	**−1.66**	**−0.74**
Mean stem cover^2	−0.13	0.21	−0.43	0.18
SD surface organic layer depth	**1.01**	**0.32**	**0.54**	**1.47**
Mean canopy cover	**0.96**	**0.27**	**0.58**	**1.34**
SD visual obstruction height	**−0.82**	**0.30**	**−1.25**	**−0.39**
SD stem cover	**0.70**	**0.24**	**0.36**	**1.05**
Mean visual obstruction height	−0.25	0.31	−0.69	0.19
Mean visual obstruction height^2	**−0.39**	**0.16**	**−0.61**	**−0.16**
Year (2023 vs. 2022)	−0.22	0.31	−0.66	0.22
Detection (P)				
Intercept	−0.59	0.10	−0.73	−0.45
Temperature	**0.57**	**0.09**	**0.44**	**0.70**
Wind	**−0.45**	**0.10**	**−0.60**	**−0.31**
Lunar phase	**−0.23**	**0.08**	**−0.34**	**−0.12**
Hour of day	0.03	0.08	−0.09	0.15
Cloud cover	0.01	0.09	−0.12	0.14

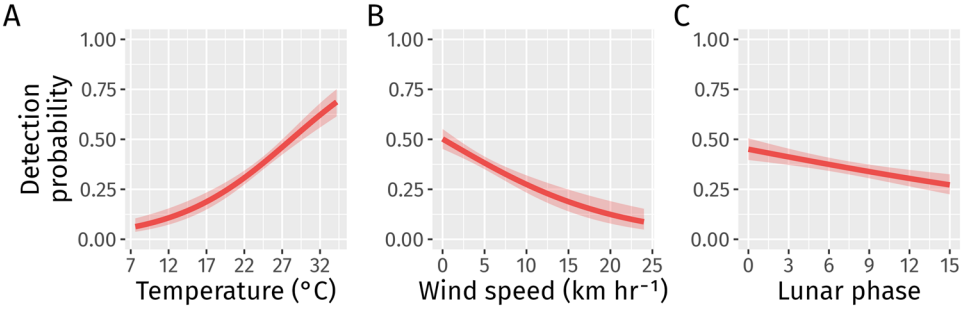


FIGURE 2. Relationships between detection probability and key environmental predictors in the best-supported occupancy model for *L. j. jamaicensis* in salt marsh habitat across the Gulf Coast of the southern U.S. Relationships are ordered from strongest to weakest, based on the magnitude of scaled parameter estimates. Detection probability increased with higher temperatures (**A**), lower wind speeds (**B**), and closer to the new moon (**C**) (0 = new moon, 15 = full moon). Shaded bands represent 85% model-estimated confidence intervals.

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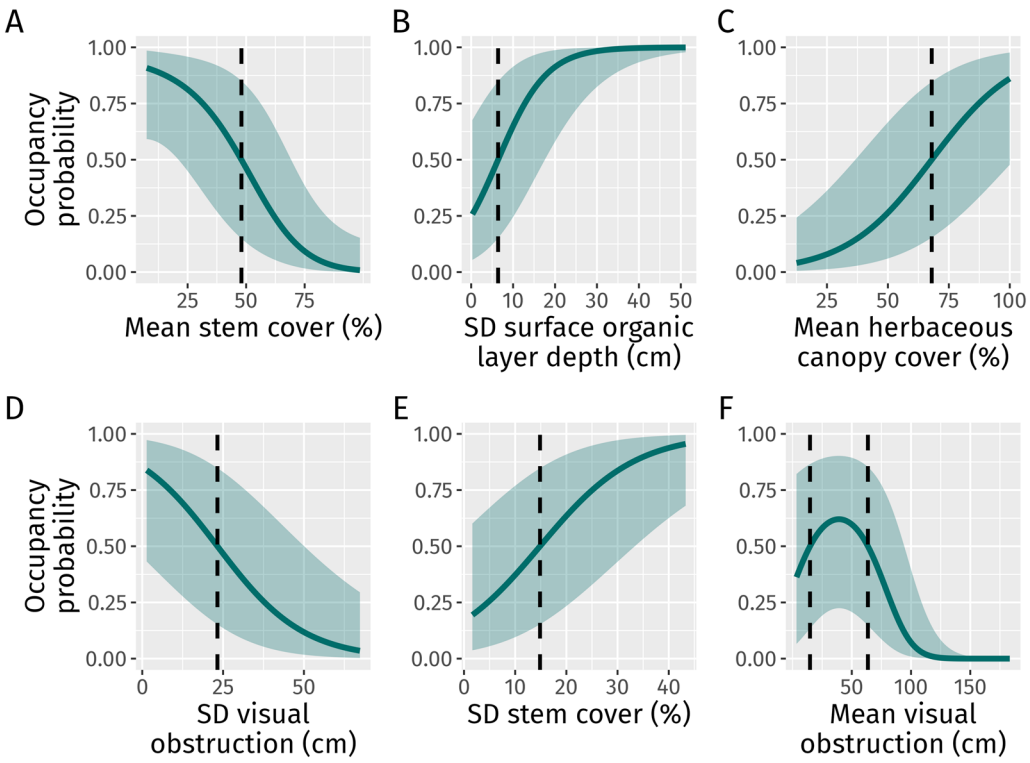


FIGURE 3. Relationships between occupancy probability and six key predictors in the best-supported model for *L. j. jamaicensis* in salt marsh habitats across the Gulf Coast of the southern U.S. Relationships are ordered from strongest to weakest, based on the magnitude of scaled parameter estimates. Occupancy probability increased at sites with lower ground-level stem cover (**A**), greater heterogeneity in surface organic layer depth (**B**), higher herbaceous canopy cover (**C**), lower heterogeneity in visual obstruction height (**D**), greater heterogeneity in ground-level stem cover (**E**), and intermediate visual obstruction height (**F**). Shaded bands represent 85% model-estimated confidence intervals. Vertical dashed lines denote the covariate values at which predicted occupancy probability equals 0.5 (i.e., the threshold at which occupancy and non-occupancy are equally likely), with all other covariates held at their mean.

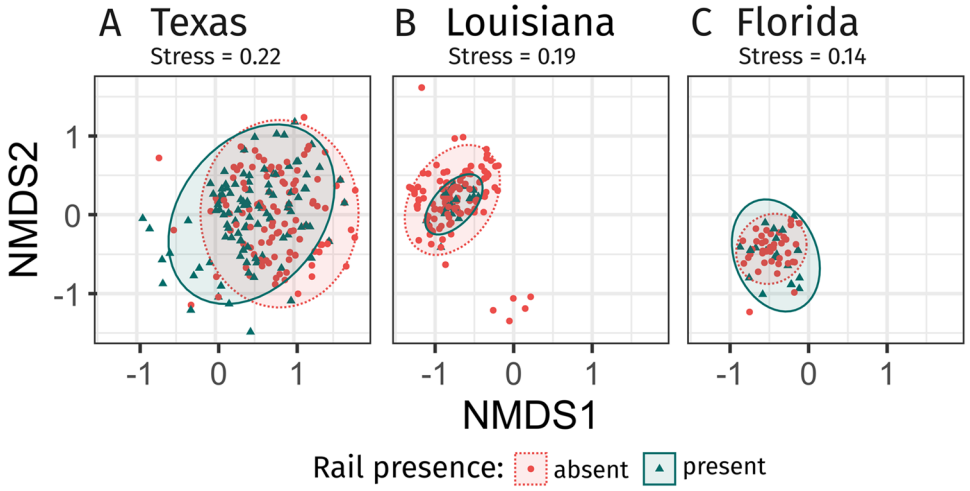


FIGURE 4. Non-metric multidimensional scaling (NMDS) ordination of site-specific vegetation species composition in coastal marshes with (triangles; solid line) and without (circles; dotted line) *L. j. jamaicensis* detections in Texas (**A**), Louisiana (**B**), and Florida (**C**). Each point represents a site, and proximity in ordination space reflects similarity in species composition. NMDS was based on Bray–Curtis dissimilarity. Stress values were 0.22, 0.19, and 0.14 for panels (**A–C**), respectively, indicating a poor fit for Texas data, but a moderate to good representation of community dissimilarity in Louisiana and Florida. Axes are held constant across panels to facilitate state-to-state comparisons.

Within states, vegetation composition differed slightly between occupied and unoccupied sites in Texas ($F_{1, 233} = 1.75$, $r^2 = 0.007$, $P < 0.001$; Figure 4A) and Louisiana ($F_{1, 166} = 1.68$, $r^2 = 0.010$, $P = 0.026$; Figure 4B), but not in Florida ($F_{1, 56} = 1.19$, $r^2 = 0.021$, $P = 0.201$; Figure 4C). However, these between-group differences were small, as suggested by low R^2 values, and were likely influenced by within-group dispersion rather than by true differences in species composition, as multivariate dispersion differed significantly between occupied and unoccupied sites in both Texas ($F_{1, 233} = 4.31$, $P = 0.034$) and Louisiana ($F_{1, 166} = 5.39$, $P = 0.017$).

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Together, these results indicate that *L. j. jamaicensis* can occupy a range of vegetation communities within and across states.

DISCUSSION

We found that *L. j. jamaicensis* occupancy was associated with dense canopy cover (>68% mean herbaceous canopy cover), an open, heterogeneous subcanopy (<48% mean ground stem cover with >15 percentage points of variation), and a heterogeneous surface organic layer (>6.4 cm of variation in depth). Occupied sites also had consistently short vegetation (16–56 cm visual obstruction height with <23 cm of variation). In contrast, there was little evidence that plant species composition varied predictably between occupied and unoccupied sites. Together, these results indicate that *L. j. jamaicensis* exhibit consistent structural habitat associations across salt marshes of the U.S. Gulf Coast, despite regional differences in plant communities, suggesting that *L. j. jamaicensis* habitat management should prioritize maintaining or creating suitable vegetation structure rather than targeting exact plant species assemblages. By identifying the habitat characteristics most strongly associated with occupancy across a broad geographic extent, this study addresses a high-priority research need for *L. j. jamaicensis* conservation (Atlantic Coast Joint Venture 2020) and provides a crucial foundation for future conservation efforts for the species.

Our work highlights the importance of vertical and understory complexity in defining high-quality *L. j. jamaicensis* habitat. Occupied sites combined a dense herbaceous canopy with a relatively open but heterogeneous subcanopy and variable surface organic layer, indicating that habitat suitability may depend on the interaction between canopy structure and ground-level conditions. Previous studies often emphasize the importance of dense herbaceous vegetation to *L. j. jamaicensis* (e.g., Watts 2022). However, focusing solely on canopy density or cover may provide an incomplete portrayal of high-quality habitat by overlooking the structural nuances that appear to be important for *L. j. jamaicensis* use (here and Butler et al. 2023). In coastal marshes, vertical complexity is often produced by plants that exhibit a growth pattern in which a tight basal clump supports stems that radiate outward, creating a closed canopy with sparse ground cover. This structure likely facilitates *L. j. jamaicensis* movement and foraging during the breeding season, when individuals rarely fly (Eddleman et al. 2020), while also providing cover from above. Understory openings and heterogeneity in the depth of the surface organic layer may also enhance habitat quality by supporting *L. j. jamaicensis* prey, as dense, continuous vegetation can reduce light availability and inhibit the development of algal films on sediment and plants, thereby negatively affecting macroinvertebrates (da Cunha Lana and Guiss 1992, Levin and Talley 2000).

Topography was not a strong predictor of *L. j. jamaicensis* occupancy in our study, possibly because we did not evaluate it at a fine enough spatial scale. *Laterallus j. jamaicensis* is thought to select sites with greater microtopographic variation, which can provide localized elevation refugia that reduce flood risk, a function that may become increasingly important given sea level rise (Atlantic Coast Joint Venture 2020). Microtopography may also contribute to pyrodiversity (i.e., heterogeneity in fire history) in coastal marshes, potentially providing additional benefits to *L. j. jamaicensis* (Stantial et al. 2023). Future research that quantifies microtopography at finer spatial resolutions and explicitly links elevation variability to flooding or fire dynamics may help clarify the extent to which microtopography influences *L. j. jamaicensis* habitat use and selection.

Across the U.S. Gulf Coast, we found that *L. j. jamaicensis* detection probability decreased around the full moon. This pattern probably does not indicate moonlight avoidance, as most of our surveys (and detections) occurred during the day. Instead, it may relate to the species' breeding phenology. Watts et al. (2021) documented the same pattern and posited that *L. j. jamaicensis* may synchronize their nesting with lunar cycles, initiating nests shortly after new moon spring tides, thereby mitigating the flood risk to nests. However, that study took place on the U.S. Atlantic Coast, where tidal amplitude is more closely tied to lunar phase than along the U.S. Gulf Coast (Rozas 1995). Higher daytime detectability near the new moon may reflect reduced diurnal activity following brighter lunar phases, when *L. j. jamaicensis* may shift their foraging to nighttime hours. This question is outside the scope of our study, which did not include nighttime surveys. However, several other waterbirds shift their foraging and activity patterns in response to lunar cycles, including *Limnodromus* spp. (dowitchers), *Calidris alpina* (Dunlin) and other small sandpipers (Dodd and Colwell 1998), *Vanellus vanellus* (Northern Lapwings; Milson 1984), and *Charadrius vociferus* (Killdeer; Eberhart-Phillips 2017), providing precedent for similar behavior in *L. j. jamaicensis*. Such shifts could also reduce encounters with diurnal predators. Given that *L. j. jamaicensis* is known to be active at night (Conway et al. 2004, Butler et al. 2023), these explanations are plausible but warrant further investigation.

Our estimates suggest that the probability of detecting a *L. j. jamaicensis* during a call-broadcast survey at a site where it is present is 43% under average survey conditions. This implies that 5 surveys would be required at a site to be at least 90% confident of detecting a *L. j. jamaicensis* if it is present, and 6 would be required to achieve 95% confidence. These results highlight that multiple site visits are important when survey objectives include confidently distinguishing true absences from non-detections. If a project aims to confirm *L. j. jamaicensis* absence at a site, or to minimize false absences under typical survey conditions, then 5–6 visits would be needed.

In conclusion, we have shown that vegetation structure is a strong predictor of *L. j. jamaicensis* occupancy in coastal marshes along the U.S. Gulf Coast. *Laterallus j. jamaicensis* was more likely to occupy habitats with dense vegetation cover and a relatively open, heterogeneous subcanopy, despite regional differences in plant species composition. However, occupancy varied substantially across the 3 U.S. states in our study region. Texas had the highest occupancy, while Louisiana had the lowest, which may reflect a combination of reduced management on Louisiana's private lands and impacts from the 2020 hurricane season. To conserve this highly imperiled species, future work should examine how to create and maintain these structural vegetation conditions through management practices such as prescribed fire. Frequent (e.g., ≤4-yr return intervals), patchy, low- to moderate-intensity burns may prevent woody encroachment, stimulate regrowth of dominant herbaceous species, and generate fine-scale heterogeneity in the subcanopy (Cox et al. 2025). However, the effects of fire are likely to depend on local site factors, including rainfall, soil

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moisture, inundation history, microtopography, and existing vegetation structure (Romano and Ursino 2020). Research examining these interactions can provide managers with practical guidance on how to implement fire to reliably create and maintain high-quality *L. j. jamaicensis* habitat. Researchers should also prioritize questions that reduce key uncertainties and directly inform management decisions, such as identifying optimal, region-specific fire return intervals for maintaining vertical complexity in coastal marshes (Stantial et al. 2023). Understanding how fire and other management practices interact with vegetation structure, hydrology, and microtopography will be crucial for designing flexible, landscape-scale strategies that support *L. j. jamaicensis* while also achieving broader conservation objectives in coastal wetlands.

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Ethics statement

This research was conducted using non-invasive methods that did not require the capture or handling of birds. Institutional Animal Care and Use Committee and Migratory Bird Permits were not required for this research.

Conflict of interest statement

The authors declare no conflict of interest.

Author contributions

E.A.B.: conceptualization, data curation, formal analysis, methodology, visualization, writing–original draft, writing—review and editing; P.A.: data curation, investigation, writing—review and editing; M.J.B.: conceptualization, data curation, investigation, methodology, writing—review and editing; W.C.C.: conceptualization, data curation, funding acquisition, investigation, methodology, resources, writing–review and editing; J.A.C.: conceptualization, funding acquisition, methodology, supervision, writing—review and editing; E.I.J.: conceptualization, investigation, funding acquisition, methodology, supervision, writing—review and editing; H.E.L.: data curation, methodology, writing—review and editing; J.T.L.: data curation, investigation, methodology, supervision; J.A.M.: conceptualization, data curation, funding acquisition, investigation, methodology, resources, writing—review and editing; M.S.W.: conceptualization, data curation, funding acquisition, investigation, methodology, project administration, resources, supervision; A.M.V.F.: conceptualization, data curation, funding acquisition, investigation, methodology, project administration, resources, supervision, writing—review and editing.

Data availability

Analyses reported in this article can be reproduced using data available from Beilke et al. (2026).

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